



=== Harwell RSX Characterization Report ===

Each resistor undergoes characterization at multiple temperature points to develop a precise model for calculating resistance across its operating range. Our modern resistors use a proprietary metal foil design that depending on the unique assembly of each resistor, we may characterize it via a polynomial or third-degree fit.

=== Instrument Summary ===

Date: JUNE 9, 2026

Model: Harwell RS-7

Value: 100000  $\Omega$

S/N: RS70297

Public Report Reference: 26225

=== Calibration Data ===

Calibrated Resistance: 99998.349  $\Omega$  at 23°C

Expanded Uncertainty of Measurement, k=2: 1  $\mu\Omega/\Omega$

Measurement Current: 50  $\mu\text{A}$

=== Temperature Characterization Data ===

$\alpha$ : -2.304E-06  $\Omega/\Omega/^\circ\text{C}$  (-2.304 ppm/ $^\circ\text{C}$ )

$\beta$ : 2.210E-07  $\Omega/\Omega/^\circ\text{C}^2$  (0.221 ppm/ $^\circ\text{C}^2$ )

Linear TCR (18-28°C): -2.22 ppm/ $^\circ\text{C}$

To calculate the resistor value at a temperature other than 23°C, use the following model where T is the desired temperature in  $^\circ\text{C}$ :

$$R(T) = R_{(23)}[1 + \alpha(T - 23) + \beta(T - 23)^2]$$

Note: This model may not be accurate outside the characterized temperature range.

| Temperature ( $^\circ\text{C}$ ) | Temperature ( $^\circ\text{F}$ ) | Resistance ( $\Omega$ ) | Deviation (ppm) |
|----------------------------------|----------------------------------|-------------------------|-----------------|
| 18.0                             | 64.4                             | 100000.053              | 17.04           |
| 18.5                             | 65.3                             | 99999.833               | 14.84           |
| 19.0                             | 66.2                             | 99999.624               | 12.75           |
| 19.5                             | 67.1                             | 99999.426               | 10.77           |
| 20.0                             | 68.0                             | 99999.239               | 8.90            |
| 20.5                             | 68.9                             | 99999.063               | 7.14            |
| 21.0                             | 69.8                             | 99998.898               | 5.49            |
| 21.5                             | 70.7                             | 99998.744               | 3.95            |
| 22.0                             | 71.6                             | 99998.601               | 2.52            |
| 22.5                             | 72.5                             | 99998.470               | 1.21            |
| 23.0                             | 73.4                             | 99998.349               | 0.00            |
| 23.5                             | 74.3                             | 99998.239               | -1.10           |
| 24.0                             | 75.2                             | 99998.141               | -2.08           |
| 24.5                             | 76.1                             | 99998.053               | -2.96           |
| 25.0                             | 77.0                             | 99997.977               | -3.72           |
| 25.5                             | 77.9                             | 99997.911               | -4.38           |
| 26.0                             | 78.8                             | 99997.857               | -4.92           |
| 26.5                             | 79.7                             | 99997.813               | -5.36           |
| 27.0                             | 80.6                             | 99997.781               | -5.68           |
| 27.5                             | 81.5                             | 99997.760               | -5.89           |
| 28.0                             | 82.4                             | 99997.749               | -6.00           |

Harwell Resistor TCR  
Model: RS-7 | Nominal Value: 100000  $\Omega$  | Serial Number: RS70297

